DBMS MCQ-UNIT -1

1. Which of the following best describes a database?
a) A collection of unrelated data
b) A structured collection of related data stored for retrieval
c) A set of tables without any relationships
d) A file system
Answer: b

2. Which of the following is a benefit of using a database system?
a) Data redundancy increases
b) Data sharing is not possible
c) Ensures data consistency
d) Higher dependency on files
Answer: c

3. Which of the following is not a characteristic of a database approach?
a) Data abstraction
b) Program-data independence
c) Multiple View
d) Dependent file formats
Answer: d

4. What does the three-schema architecture help achieve?
a) Data redundancy
b) Data independence
c) Faster database access
d) Improved file handling
Answer: b

5. Which level of the three-schema architecture provides a user's view of the database?
a) Internal level
b) Conceptual level
c) External level
d) Physical level
Answer: c

6. Which type of independence is achieved when changes in the physical schema do not affect the logical schema?
a) Logical independence
b) Physical independence
c) Data independence

d) None of the above
Answer: b

7. In ER-to-Relational mapping, what does each entity type translate into?
a) A relationship
b) A foreign key
c) A table
d) A schema
Answer: c

8. What does a multi-valued attribute in ER model translate to in a relational model?
a) A single table column
b) A separate table
c) A foreign key
d) A derived attribute
Answer: b

9. In a database system, what is meant by "concurrency control"?
a) Preventing data redundancy
b) Ensuring simultaneous access to data without conflicts
c) Backing up data regularly
d) Controlling user permissions
Answer: b

10. Which key is used to establish a relationship between two tables in a relational model?
a) Primary key
b) Foreign key
c) Composite key
d) Surrogate key
Answer: b

11. What does an ER diagram represent?
a) Relationships between files in a database
b) Data and relationships in a database
c) Hardware architecture of a database
d) The software used in a database
Answer: b

12. Which type of attribute cannot be divided into smaller parts?
a) Composite attribute
b) Derived attribute
c) Multivalued attribute

d) Simple attribute
Answer: d

13. How are multi-valued attributes represented in an ER diagram?
a) Ellipse
b) Double ellipse
c) Dashed ellipse
d) Rectangle
Answer: b

14. Which of the following attributes is a derived attribute?
a) Date of birth
b) Age
c) Name
d) Address
Answer: b

15. Which attribute in an ER diagram is unique for every entity?
a) Foreign key
b) Primary key
c) Composite attribute
d) Multi-valued attribute
Answer: b

16. Which of the following is an example of a many-to-many relationship?
a) A teacher teaches one class
b) A student enrolls in multiple courses, and a course has multiple students
c) A customer buys one product
d) A manager supervises one employee
Answer: b

17. Which symbol is typically used to represent an entity in an ER diagram?

a. Circle
b. Square
c. Rectangle
d. Triangle

Ans: (c)

18. What is a weak entity in an ER model?

a. An entity that does not have a key attribute
b. An entity that cannot exist without a relationship with another entity
c. An entity that has multiple key attributes
d. An entity that has no attributes

Ans: (a)

19.In an ER model, what does a diamond symbol represent?

a. Attribute
b. Entity
c. Relationship
d. Primary Key

Ans: (c)

20. Which of the following is NOT a type of attribute in an ER model?

a. Simple attribute
b. Composite attribute
c. Derived attribute
d. Aggregate attribute

Ans: (d)

21.Which of the following best describes a multi-valued attribute?

a. An attribute that can have multiple values at the same time
b. An attribute that is derived from other attributes
c. An attribute that is always numeric
d. An attribute that has a unique value for each entity

Ans: (a)

22. What is cardinality in the context of an ER model?

a. The number of entities involved in a relationship
b. The number of attributes an entity has
c. The number of relationships an entity has
d. The number of entities in a database

Ans: (a)

23. Which of the following is true about a primary key in an ER model?

a. It uniquely identifies each entity in an entity set
b. It can have duplicate values
c. It can be NULL
d. It can be a combination of multiple attributes

Ans: (a)

24. A relationship where an entity is related to itself is called:

a. Binary Relationship
b. Recursive Relationship
c. Ternary Relationship
d. Self-Relationship

Ans: (b)

25. What does an ellipse symbolize in an ER diagram?

a. Entity
b. Attribute
c. Relationship
d. Weak Entity

Ans: (b)

26. Which type of attribute cannot be further divided?

a. Derived Attribute
b. Composite Attribute
c. Simple Attribute
d. Multivalued Attribute

Ans: (c)

27. Which one of the following uniquely identifies the elements in the relation?
a) Secondary Key
b) Primary key
c) Foreign key
d) Composite key
Ans: b

28. ______ is a special type of integrity constraint that relates two relations & maintains consistency across the relations.
a) Entity Integrity Constraints
b) Referential Integrity Constraints
c) Domain Integrity Constraints
d) Domain Constraints
Ans: b

29. Foreign key is the one in which the **________** of one relation is referenced by another relation.
a) Foreign key
b) Primary key
c) References
d) Check constraint
Ans: b

30. For each attribute of a relation, there is a set of permitted values, called the ________ of that attribute.
a) Domain
b) Relation

c) Set
d) Schema
Ans: a

31. The tuples of the relations can be of **________** order.
a) Any
b) Same
c) Sorted
d) Constant
Ans: a

32.Let X(a, b, c, d) and Y(p,q r, s) be two relations in which a is foreign key of X that refers to primary key of Y. Which of the following operation causes referential integrity constraint violation?
a) insert into Y
b) Delete from X
c)insert into X
d) All above
Ans: c

33. Which of the following operation do not violates domain constraints?
a) Insert operation
b) Delete operation
c) Update operation
d) All above
Ans: b

34. **Which one of the following is a set of one or more attributes taken collectively to uniquely identify a record?**
a) Candidate key
b) Sub key
c) Super key
d) Foreign key
Ans: c

35. **Which of the following is a correct statement regarding a relational schema?**

A) It defines the structure of data and the relationships between tables.
B) It only stores data, not metadata.
C) It refers to the actual data stored in the database.
D) It cannot include any constraints.

Answer: A

36.**Which of the following is a violation of a referential integrity constraint?**
A) Attempting to insert a duplicate value in a primary key column
B) Inserting a value into a foreign key column that does not exist in the primary key column of the referenced table
C) Updating a value in a foreign key column
D) All of the above
Answer: B

37.**Which of the following statements is true about a primary key constraint?**

A) It allows duplicate values but ensures that each value is unique within the table.
B) It ensures that no two rows in a table can have the same value for the primary key attribute(s).
C) It can be null.
D) It can be created on any non-key column.

Answer: B

38.**Which of the following operations are part of relational update operations?**
A) Insert
B) Delete
C) Update
D) All of the above
Answer: D

39.**Which of the following is NOT a fundamental concept of the Relational Model?**
A) Relations
B) Tuples
C) Keys
D) Objects
Answer: D

40.The relational algebra query language is ___.

a. Analytical
b. Procedural
c. Symmetrical
d. Instrumental

Answer: B

41. In Relational Algebra, queries are performed using ___.

a. Entities
b. Relationships
c. Operators
d. Objects

Answer: C

42. For select operation the ________ appear in the subscript and the ___________ argument appears in the parenthesis after the sigma.

a. Predicates, relation
b. Relation, Predicates
c. Operation, Predicates
d. Relation, Operation

Answer: A

43. The select, project and rename operations are called____

a. Binary operations
b. Ternary Operations
c. Unary operations
d. None of the above.

Answer: c

44. If in the JOIN operation, the conditions of JOIN operation are not satisfied then the results of the operation is

a. 0
b. 1
c. 2
d. 3

Answer: a

45.The DIVISION operation can be applied to two relations A and B such as R(A) ÷ R(B) where as

a. A does not belong to B
b. A must be added to B
c. A belongs or equal to B
d. A must be subtracted from B

Answer: c

46. The operation which allows to process relationships from multiple relations rather than single relation is classified as

a. division operation
b. relation operation
c. square operation
d. join operation

Answer: d

47. The kind of operation by which one tuple can be created with the help of combining tuples from two relations is classified as

a. square operation
b. join operation
c. division operation
d. relation operation

Answer: B

48. The ________ clause is used to list the attributes desired in the result of a query.

a. Where
b. Select
c. From
d. Distinct

Answer: B

49. Which of the following statements contains an error?

a. Select * from emp where empid = 10003;
b. Select empid from emp where empid = 10006;
c. Select empid from emp;
d. Select empid where empid = 1009 and lastname = ‘GELLER’;

Answer: D

50. **Which of the following is a key concept of the Relational Model?**

A) Data is stored in hierarchical structure
B) Data is organized in tables (relations)
C) Data is stored in linked lists
D) Data is represented as objects

Answer: B